

Solving and Graphing One-Step Inequalities

One-step inequalities: undo an addition, undo a multiplication, and turn the sign around when the divisor is negative

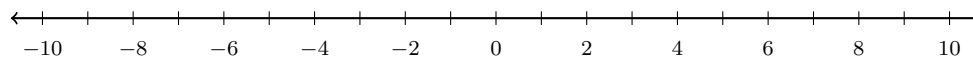
Directions:

- **Worked pattern.** To solve $x - 4 > 3$, add 4 to both sides: $x > 7$. Graph it with an *open* circle at 7 (because $>$ does not include 7) and shade to the right.
- Solve each inequality in one step. Write the solution in the form “letter, sign, number” on the answer line, then graph it.
- **Open or filled?** $<$ and $>$ get an open circle; $\leq$ and $\geq$ get a filled one.
- **Turn the sign around** only when you multiply or divide both sides by a *negative* number — never because the answer happens to be negative.

1. Solve and graph: $x + 5 < 12$

(a) the solution: _____

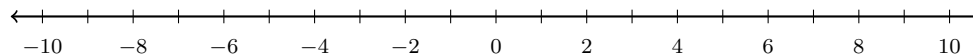
(b) graph the solution on the number line:



2. Solve and graph: $y - 3 \geq 4$

(a) the solution: _____

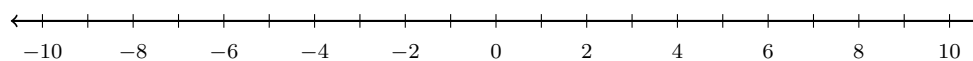
(b) graph the solution on the number line:



3. Solve and graph: $4m \leq 20$

(a) the solution: _____

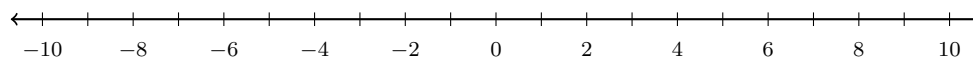
(b) graph the solution on the number line:



4. Solve and graph: $n + 9 > 2$

(a) the solution: _____

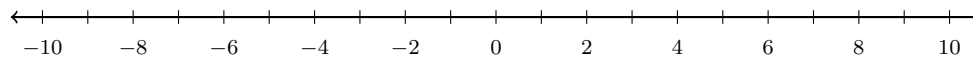
(b) graph the solution on the number line:



5. Solve and graph: $\frac{p}{3} > 2$

(a) the solution: _____

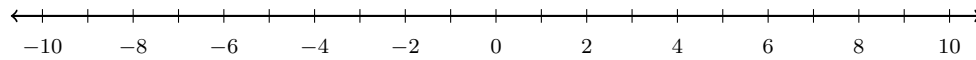
(b) graph the solution on the number line:



6. Solve and graph: $-2x \geq 10$

(a) the solution: _____

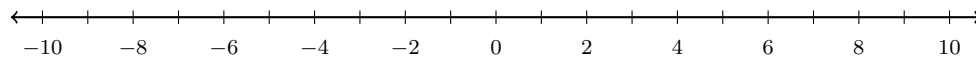
(b) graph the solution on the number line:



7. Solve and graph: $t - 6 \leq -1$

(a) the solution: _____

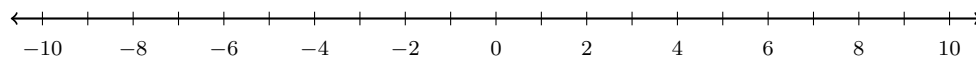
(b) graph the solution on the number line:



8. Solve and graph: $7w < -21$

(a) the solution: _____

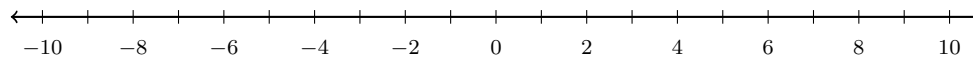
(b) graph the solution on the number line:



9. Solve and graph: $-\frac{k}{4} < 2$

(a) the solution: _____

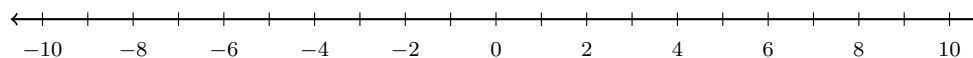
(b) graph the solution on the number line:



10. Solve and graph: $15 \geq 3r$

(a) the solution: _____

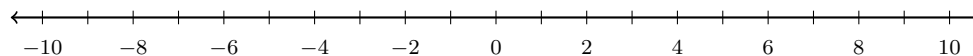
(b) graph the solution on the number line:



11. Solve and graph: $-5s < 35$

(a) the solution: _____

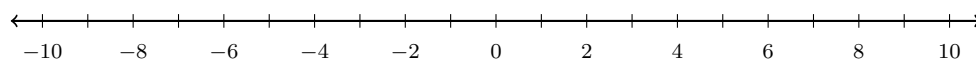
(b) graph the solution on the number line:



12. Solve and graph: $-3v \geq 18$

(a) the solution: _____

(b) graph the solution on the number line:



(c) Explain why the sign turns around in this problem, and back it up by testing one number from the shaded part of your graph in the original inequality. Write this in the work space.